

Aditya Keshavapatnam

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Education

University of North Texas, Masters in Advanced Data Analytics

Aug 2023 – Dec 2024

Work Experience

Data Scientist - Sports Analytics, Mean Green Athletics – Denton, TX

March 2025 – Present

- Managed Catapult GPS technology and OpenField software, overseeing the end-to-end data lifecycle, from collection and validation to analysis and archival for more than 100+ athletes.
- Directed real-time data operations, ensuring accurate tracking of over 50 physical performance metrics, including total distance, high-speed running, acceleration load, and player load during training and games.
- Delivered daily performance visuals and athlete rankings, improving athlete engagement and coaching decisions; reduced data turnaround time by 40%.
- Engineered interactive dashboards in Power BI, automating key performance indicators (KPIs) for strength, speed, and workload metrics, enhancing daily decision-making and saving 5+ hours weekly.
- Analyzed high-volume and noisy sensor data from wearable GPS units to identify key biomechanical and workload indicators linked to injury risk, contributing to a 25% reduction in soft-tissue injuries through proactive training adjustments and recovery strategies.
- Utilized time series analysis in Python to process and interpret high-frequency sensor data, identifying inefficiencies in tackling mechanics and sharp directional cuts; implemented insights that led to a 30% improvement in movement efficiency through personalized technique optimization.

Research Assistant - NLP & Data Analytics, University of North Texas – Denton, TX

Dec 2024 – Present

- Applied Natural Language Processing (NLP) and Predictive Analytics to analyze patient statements and assess the effectiveness of different art therapy modalities in treating mental health conditions arising from trauma and stress exposure to support research in healthcare.
- Managing and maintaining research healthcare data systems, performing daily updates to ensure seamless integration, data accuracy, and compliance with research policies.
- Provided academic research support, by assisting departmental professor with reviewing and publication of research papers.

Graduate Teaching Assistant, University of North Texas – Denton, TX

Jan 2024 – Dec 2024

- Mentored more than 200 graduate students in Python, SAS, GCP, Power BI, and Tableau, fostering hands-on expertise in advanced data analytics and driving a 95% assignment completion rate.
- Analyzed student performance data to identify areas for improvement, providing customized feedback, and ensuring consistent progress in academic and technical skills.
- Collaborated with faculty to design and implement course materials, aligning instructional content with industry trends and program objectives.
- Prepared and delivered interactive lab sessions and technical presentations, enhancing student engagement and achieving high satisfaction ratings.

Student Assistant - Data Collection & Assessment, University of North Texas –
Denton, TX

Aug 2023 – Dec 2023

- Cleaned and prepared large datasets for analysis using Python and Excel, including removing incomplete records, correcting date inconsistencies, applying conditional formatting, and standardizing numeric formats to ensure high data quality and reliability.
- Developed visual reports and summary tables, including time-based usage trends and platform-specific breakdowns, to support decision-making and improve the accessibility of insights for library staff and stakeholders.

Data Projects

Decoding NFL Injuries: Risk Factors, Recurrence, and Predictive Insights Aug 2024 - Nov 2024

- Analyzed 27,921 NFL injury records, identifying key risk factors such as defensive backs' 5,000+ injuries, 1.42x higher risk on artificial turf, and a 65% recurrence rate for lower leg injuries.
- Achieved 79% accuracy with XGBoost (F1: 0.64), highlighting BMI (25-29.9), age (23-29), and prior injuries as drivers of 65% of injuries.

Impacts of Infrastructure and Pre-Existing Health Conditions impacts during COVID-19 Period May 2024 - July 2024

- Examined 406,170 COVID-19 records to build predictive models explaining 99.6% of death variability, identifying vaccination rates, ICU availability, and pre-existing conditions as key predictors.
- Highlighted infrastructure and lifestyle factors like diabetes and smoking on COVID-19 severity, revealing a 59% drop in U.S. cases among older populations post-vaccination.

Investment Dynamics: Exploring Returns and Risks in Cryptocurrencies and Tech Stocks Jan 2024 - April 2024

- Analyzed 75 months of historical data, identifying Binance Coin (7,100%) and Nvidia (1,512%) as top performers, with BNB-USD showing the highest volatility ($\beta = 14.97$, $SD = 21.5$) compared to NVDA ($\beta = 10.73$, $SD = 3.60$).
- Acquired tech stock and cryptocurrency data via Yahoo Finance API, calculated cumulative returns against benchmarks, and developed a high-accuracy LSTM model for multi-day return forecasting.

Analysis of the football Team Performance within the ACC conference from 2009 – 2018 Jan 2024 - April 2024

- Engineered a data pipeline analyzing historical data across 11 metrics to identify the top 5 performing teams, integrating supplementary datasets to address team conference changes.
- Built an end-to-end ETL process in Python for 180+ files, consolidating player-level data into NCAA-compliant team-level insights, optimizing datasets, and deploying a Tableau dashboard for strategic funding decisions.

Technical Skills

Programming Languages: Python ,PySpark, R, SQL, Java, C, C++ , NoSQL, Shell Scripting

Softwares & Cloud Services : SAS, SPSS, RStudio, Microsoft Excel, Power Automate Google Cloud Platform (Big Query, Dataflow,Cloud Storage, Cloud Functions,Dataproc,Vertex AI, Databricks)

Machine Learning: TensorFlow, scikit-learn, PyTorch, OpenCV, Ultralytics, A/B Testing, Statistics, dmba, SciPy,Numpy,Pandas

Deep Learning: Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN)

Data Engineering & Database Management: ETL, ELT, OpenRefine, MySQL, Microsoft SQL Server, Microsoft Access, Big Query, PostgreSQL, Cloud SQL, Cloud Spanner,BigTable, Firestore, HBase, Apache Airflow, Apache Nifi, Apache Kafka

Data Visualization : Matplotlib, Plotly, ggplot, Seaborn, Tableau, Power BI, Looker

CI/CD & Version Control : Cloud Build, Cloud Run, Artifact Registry, Docker, Git

Certifications

Associate Data Practitioner, Google Cloud

Data Engineer Associate, Datacamp

Develop GenAI Apps with Gemini and Streamlit Skill Badge, Google

Create Generative AI Apps on Google Cloud, Google

Create ML Models with BigQuery ML Skill Badge, Google

Prepare Data for ML APIs on Google Cloud Skill Badge, Google

Engineer Data for Predictive Modeling with BigQuery ML Skill Badge, Google